

How to Create Cloud Watch and monitor our system in AWS

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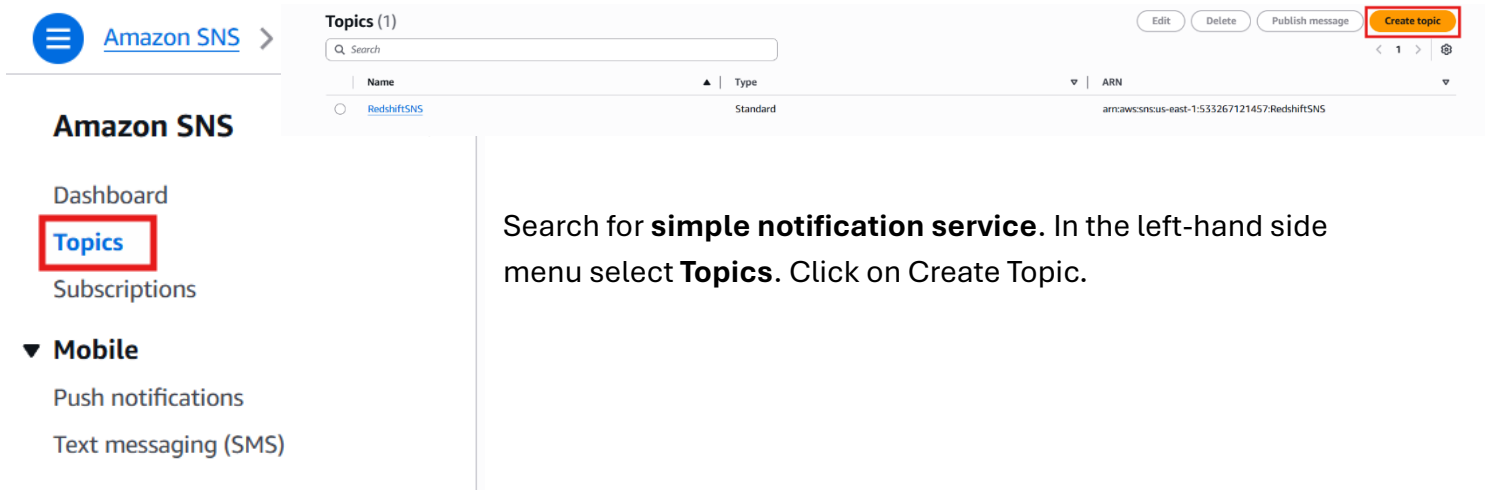
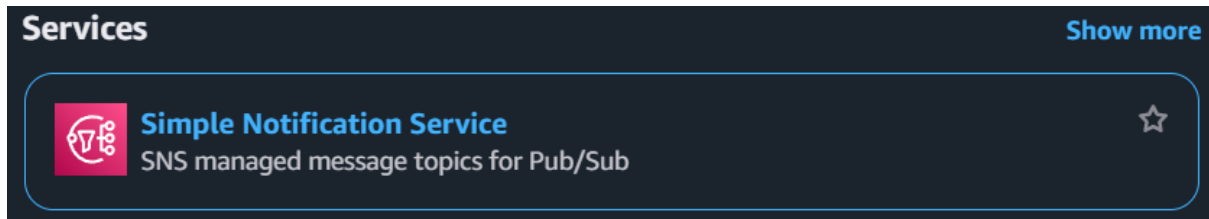
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Note: This topic will explain how to set up alarm and monitor your AWS EC2 instances.

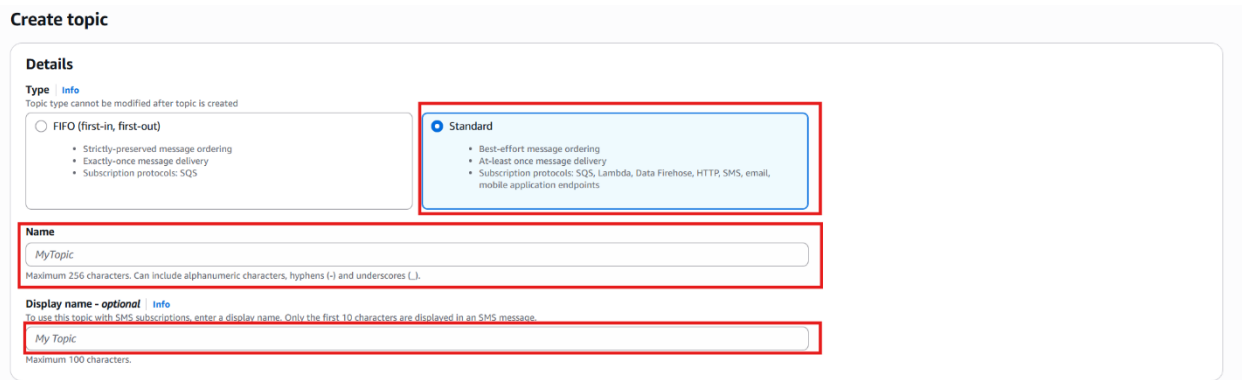
This is not included in the Final Exam only for the project and your own knowledge.

Requirements: you must have one EC2 instance running.

How to Create an simple notification service SNS

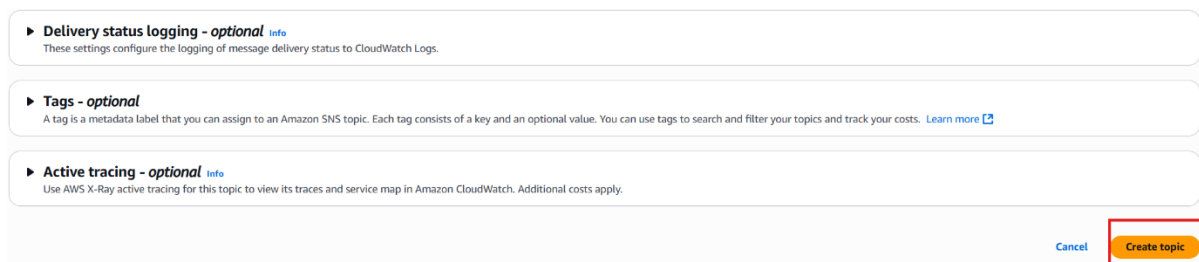


Search for **simple notification service**. In the left-hand side menu select **Topics**. Click on Create Topic.



The screenshot shows the 'Create topic' form in the AWS console. The 'Details' section is expanded, showing the 'Type' dropdown menu. The 'Standard' option is selected and highlighted with a red box. Below this, the 'Name' field is highlighted with a red box, containing the text 'MyTopic'. The 'Display name - optional' field is also highlighted with a red box, containing the text 'My Topic'.

First, choose **standard**. Second, enter a name for your topic you can give a Display name if you want.



The screenshot shows the 'Create topic' form with optional settings expanded. The 'Delivery status logging - optional' section is expanded, showing a checkbox for 'Delivery status logging' and a description. The 'Tags - optional' section is expanded, showing a checkbox for 'Tags' and a description. The 'Active tracing - optional' section is expanded, showing a checkbox for 'Active tracing' and a description. The 'Create topic' button is highlighted with a red box.

Leave the rest of the options as the default and click **Create topic**.

Subscriptions (0)

Search

ID Endpoint Status Protocol

No subscriptions found
You don't have any subscriptions to this topic.

Create subscription

under your topic you have created it will ask you to **create subscription** click on it.

Create subscription

Details

Topic ARN
arn:aws:sns:us-east-1:533267121457:Hussain

Protocol
The type of endpoint to subscribe
Email

Endpoint
An email address that can receive notifications from Amazon SNS.
test@example.com

After your subscription is created, you must confirm it. [info](#)

Subscription filter policy - optional [info](#)
This policy filters the messages that a subscriber receives.

Redrive policy (dead-letter queue) - optional [info](#)
Send undeliverable messages to a dead-letter queue.

Cancel Create subscription

First, you will notice that automatically it chooses the topic that you created. Second, for the protocol choose E-mail then enter your email that you want AWS to send you the notification on it. Last click on **Create subscription**.

Subscription: 1ff70676-aa4e-4dcb-899e-63d7d9e613e7

Edit Delete

Details

ARN
arn:aws:sns:us-east-1:533267121457:Hussain:1ff70676-aa4e-4dcb-899e-63d7d9e613e7

Endpoint
202202674@student.polytechnic.bh

Topic
[Hussain](#)

Subscription Principal
arn:aws:iam::533267121457:role/voclabs

Status
Pending confirmation

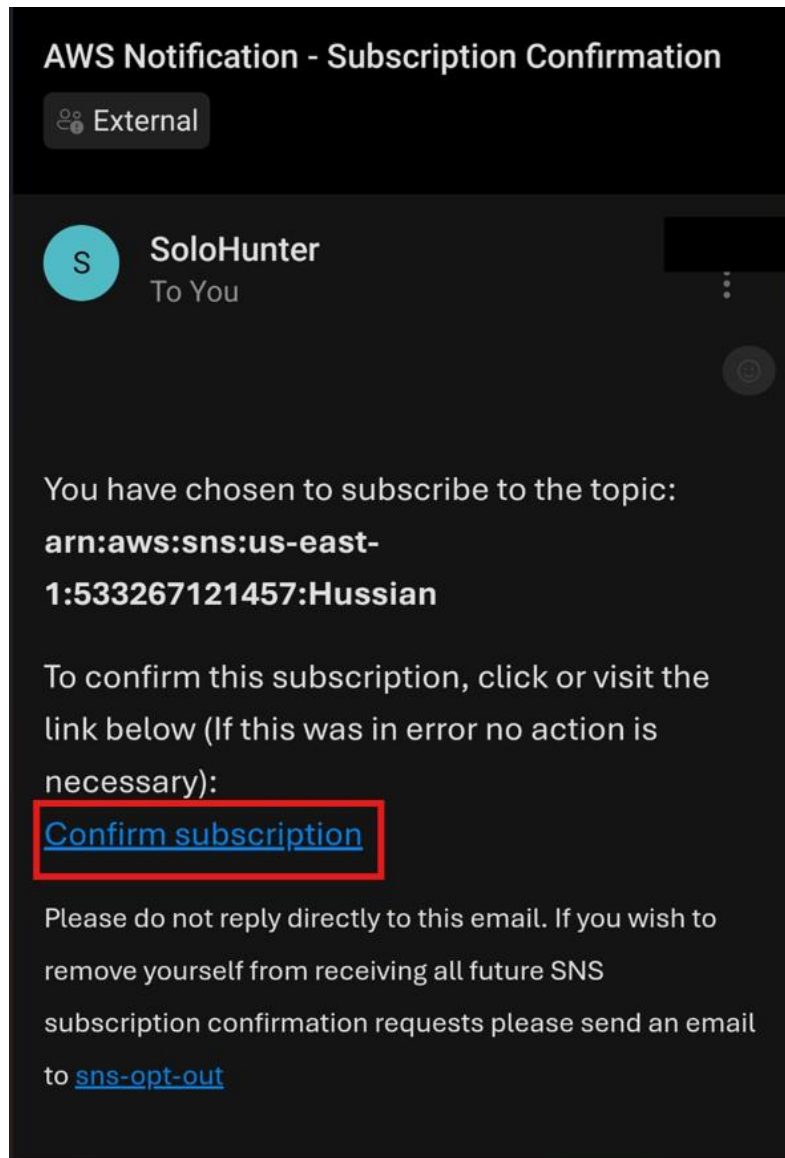
Protocol
EMAIL

Subscription filter policy [info](#)
This policy filters the messages that a subscriber receives.

No filter policy configured for this subscription.
To apply a filter policy, edit this subscription.

Edit

After you create the subscription you will see that in the **Status** says pending confirmation. Go to your E-mail address that you enter you will have received an E-mail from AWS to confirm the **Subscription**.



Click on **Confirm subscription**.

Subscription: 1ff70676-aa4e-4dcb-899e-63d7d9e613e7

Edit

Delete

Details

ARN

arn:aws:sns:us-east-1:533267121457:Hussain:1ff70676-aa4e-4dcb-899e-63d7d9e613e7

Endpoint

202202674@student.polytechnic.bh

Topic

Hussain

Subscription Principal

arn:aws:iam::533267121457:role/voclabs

Status

Confirmed

Protocol

EMAIL

Subscription filter policy

Redrive policy (dead-letter queue)

Subscription filter policy [Info](#)

This policy filters the messages that a subscriber receives.

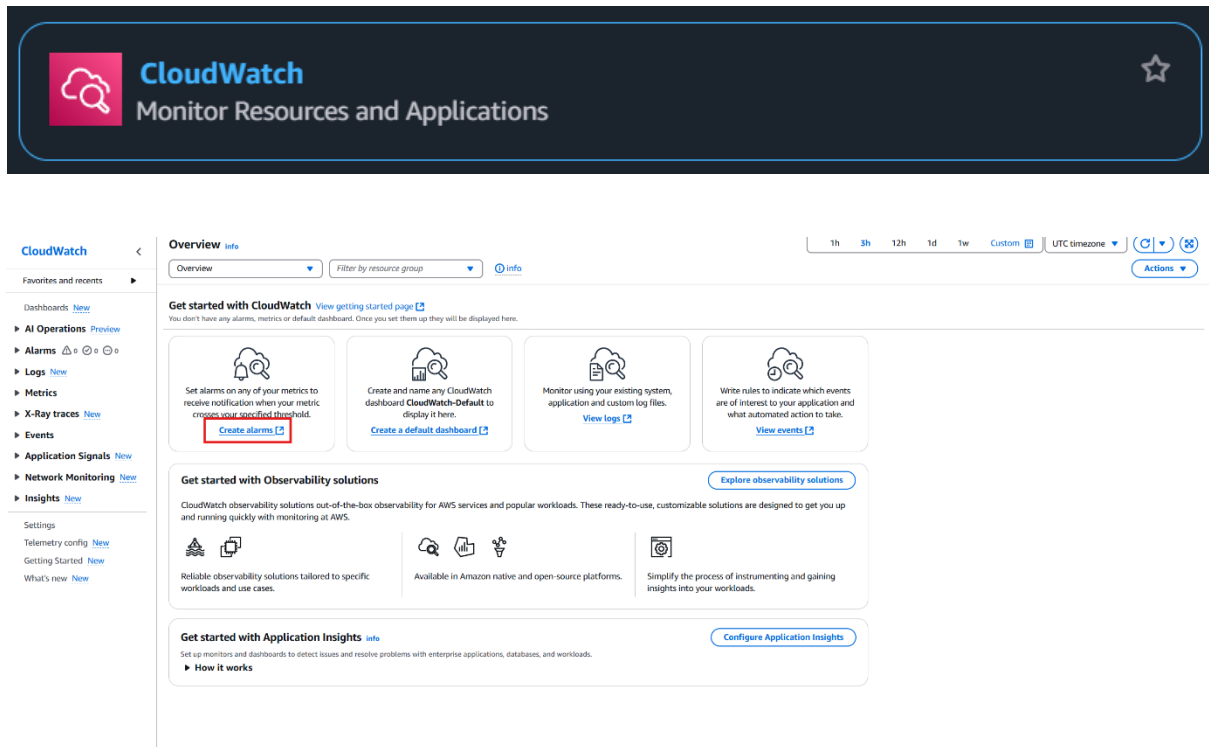
No filter policy configured for this subscription.

To apply a filter policy, edit this subscription.

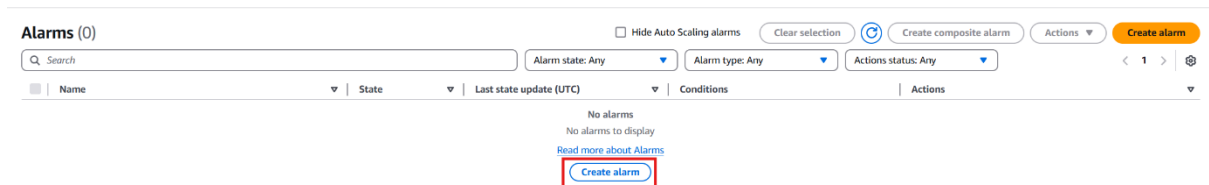
Edit

After you confirm the E-mail, you will see the status of the subscription changed to **confirmed**.

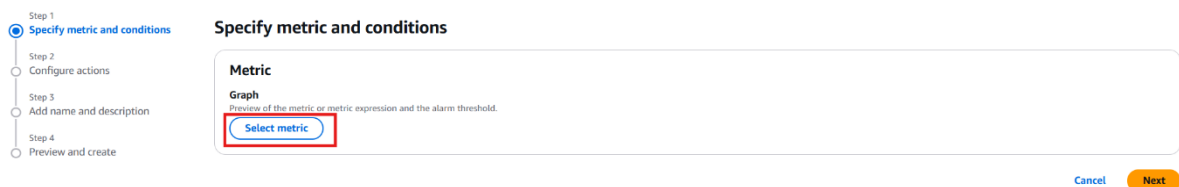
How to Create alarms



Search for cloud Watch the click on create alarms



Click on **Create alarm**



In step 1 click on **select metric**

Select metric ✕

Untitled graph [🔗](#) 1h 3h 12h 1d 3d 1w Custom UTC timezone Line 🔄

1
0.5
0

Your CloudWatch graph is empty.
Select some metrics to appear here.

21:00 21:15 21:30 21:45 22:00 22:15 22:30 22:45 23:00 23:15 23:30 23:45

[Browse](#) [Multi source query](#) [Graphed metrics](#) [Options](#) [Source](#) = [Add math](#) [Add query](#)

Metrics (1,351) 🔔 Alarm recommendations [Graph with SQL](#) [Graph search](#)

[N. Virginia](#)

ApplicationELB	65	Billing	13	EBS	217	EC2	510
Events	21	Lambda	26	Logs	6	NATGateway	80
RDS	120	Rekognition	6	SNS	6	Usage	281

[Cancel](#) [Select a single metric to continue](#)

click on **EC2**

Select metric ✕

Untitled graph [🔗](#) 1h 3h 12h 1d 3d 1w Custom UTC timezone Line 🔄

1
0.5
0

Your CloudWatch graph is empty.
Select some metrics to appear here.

21:00 21:15 21:30 21:45 22:00 22:15 22:30 22:45 23:00 23:15 23:30 23:45

[Browse](#) [Multi source query](#) [Graphed metrics](#) [Options](#) [Source](#) = [Add math](#) [Add query](#)

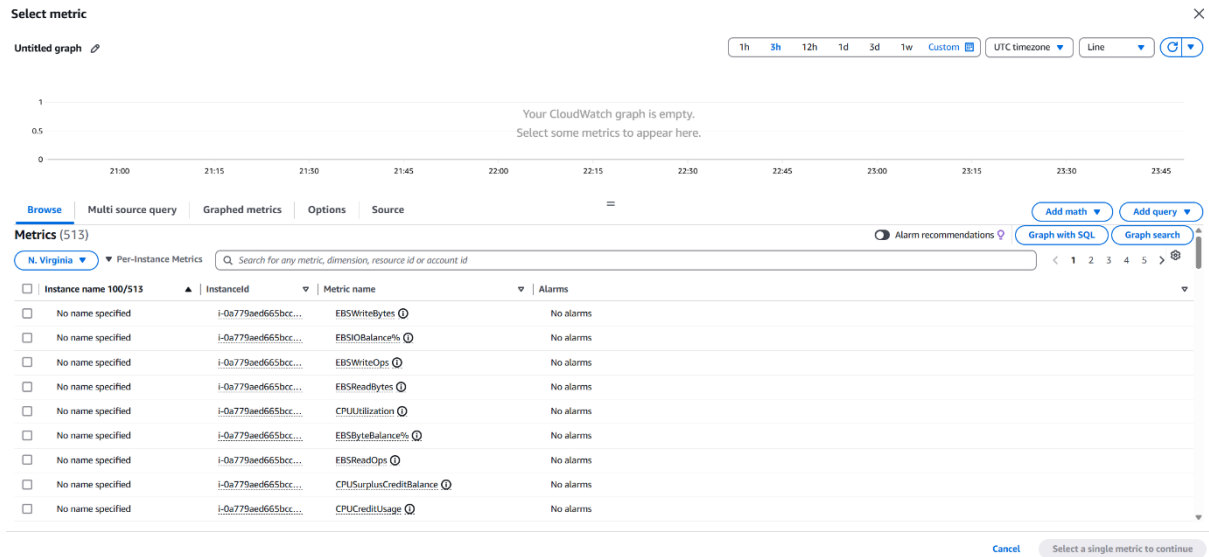
Metrics (510) 🔔 Alarm recommendations [Graph with SQL](#) [Graph search](#)

[N. Virginia](#) ▼ [EC2](#)

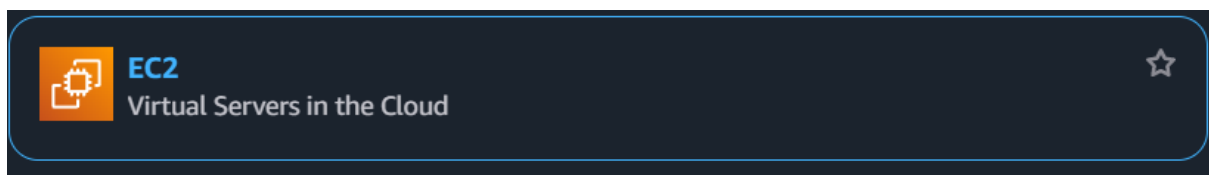
Per-Instance Metrics	510
-----------------------------	------------

[Cancel](#) [Select a single metric to continue](#)

click on **Per-Instance Metrics**



There are many EC2 instances, but you want to monitor the one you created earlier.



You can change your default landing page for EC2. Permanently dismiss Change landing page

Resources EC2 Global View

You are using the following Amazon EC2 resources in the United States (N. Virginia) Region:

Instances (running)	1	Auto Scaling Groups	0	Capacity Reservations	0
Dedicated Hosts	0	Elastic IPs	0	Instances	1
Key pairs	2	Load balancers	0	Placement groups	0
Security groups	2	Snapshots	1	Volumes	1

Account attributes

Default VPC
vpc-0f76a9bd63d923656

Settings
Data protection and security
Allowed AMIs
Zones
EC2 Serial Console
Default credit specification
EC2 console preferences

Launch instance Service health AWS Health Dashboard

Instances (1/1) Find Instance by attribute or tag (case-sensitive) All states Last updated less than a minute ago Connect Instance state Actions Launch instances

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
webserver	i-0a7e2dfee9b71a33	Running	t2.micro	2/2 checks pass	View alarms	us-east-1a	ec2-44-202-88-20.com...	44.202.88.20	-	-

i-0a7e2dfee9b71a33 (webserver)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary

Instance ID
i-0a7e2dfee9b71a33

IPv6 address
-

Hostname type
IP name: ip-172-31-90-37.ec2.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
44.202.88.20 [Public IP]

IAM Role
-

IMDSv2
Required

Operator
-

Public IPv4 address
44.202.88.20 [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-90-37.ec2.internal

Instance type
t2.micro

VPC ID
vpc-0f76a9bd63d923656

Subnet ID
subnet-06bea5d5e91af1a4

Instance ARN
arn:aws:ec2:us-east-1:533267121457:instance/i-0a7e2dfee9b71a33

Private IPv4 addresses
172.31.90.37

Public IPv4 DNS
ec2-44-202-88-20.compute-1.amazonaws.com [open address](#)

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. [Learn more](#)

Auto Scaling Group name
-

Managed
false

First, go to **EC2**. Second, choose **instance (running)**. Third, **select your EC2 instance** and from the down menu **copy your instance ID**.

Select metric

Untitled graph

1h 3h 12h 1d 3d 1w Custom UTC timezone Line

Percent

7.35

5.48

3.59

21:00 21:15 21:30 21:45 22:00 22:15 22:30 22:45 23:00 23:15 23:30 23:45

CPUUtilization

Browse Multi source query Graphed metrics (1) Options Source

Metrics (19)

N. Virginia

Per-Instance Metrics

Search for any metric, dimension, resource id or account id

i-0af7e2dfee9b71a33

Instance name 19/19	Instanceid	Metric name	Alarms
☒ webserver	i-0af7e2dfee9b71...	CPUUtilization	No alarms
☐ webserver	i-0af7e2dfee9b71...	MetadataNoToken	No alarms
☐ webserver	i-0af7e2dfee9b71...	EBSByteBalance%	No alarms
☐ webserver	i-0af7e2dfee9b71...	EBSWriteBytes	No alarms
☐ webserver	i-0af7e2dfee9b71...	EBSWriteOps	No alarms
☐ webserver	i-0af7e2dfee9b71...	EBSIOBalance%	No alarms
☐ webserver	i-0af7e2dfee9b71...	EBSReadOps	No alarms
☐ webserver	i-0af7e2dfee9b71...	EBSReadBytes	No alarms

Cancel Select metric

go back to select metric and paste the instance ID in the search bar and press enter then select **CPU Utilization** and click **select metric**.

Documentation Alarm Create alarm

Step 1: Specify metric and conditions

Step 2: Configure actions

Step 3: Add name and description

Step 4: Preview and create

Specify metric and conditions

Metric

Graph

The alarm will trigger when the blue line goes below the red line for 5 datapoints within 5 minutes.

Percent

7.35

5.48

3.59

21:00 21:15 21:30 21:45 22:00 22:15 22:30 22:45 23:00 23:15 23:30 23:45

CPUUtilization

Alarm name: AWS/EC2

Metric name: CPUUtilization

Instanceid: i-0af7e2dfee9b71a33

Instance name: webserver

Statistic: Average

Period: 5 minutes

Conditions

Threshold type: Basic

Whenever CPUUtilization is...

Greater than threshold

Greater/Equal to threshold

Lower than threshold

Lower/Equal to threshold

When... Define the threshold value: 10

Additional configuration

Cancel Next

What you have done is that the alarm will check after 5 minutes if the CPU Utilization is lower than 10 then Select **Next**

Configure actions

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ **Select an existing SNS topic**

☐ Create new topic

☐ Use topic ARN to notify other accounts

Send a notification to...

Hussian

RedshiftSNS

Add notification

For the notification select your SNS that you created before.

Note: that you will send you an email with the alarm you configured.

EC2 action

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Take the following action...
Define what will happen to the EC2 instance with the Instance ID i-0af7e2dfee9b71a33 when this alarm is triggered.

☐ Recover this instance
You can only recover certain EC2 instance types. [See documentation](#)

☒ **Stop this instance**
You can only stop an instance if it is backed by an EBS volume. AWS will use the existing Service Linked Role (AWSServiceRoleForCloudWatchEvents) to perform this action. [Show IAM policy document](#)

☐ Terminate this instance
You will not be able to terminate this instance if termination protection is enabled. AWS will use the existing Service Linked Role (AWSServiceRoleForCloudWatchEvents) to perform this action. [Show IAM policy document](#)

☐ Reboot this instance
An instance reboot is equivalent to an operating system reboot. AWS will use the existing Service Linked Role (AWSServiceRoleForCloudWatchEvents) to perform this action. [Show IAM policy document](#)

Add EC2 action

Under the EC2 action you will select stop the instance.

Systems Manager action [Info](#)

This action will create an Incident or Opsitem in Systems Manager when the alarm is **In alarm** state.

Add Systems Manager action

Investigation action - *Preview* [Info](#)

This action will create an Investigation when the alarm is **In alarm** state.

Add Investigation action

Cancel Previous **Next**

Select **next**.

Add name and description

Name and description

Alarm name

Alarm description - optional [View formatting guidelines](#)

Edit

Preview

```
# This is an H1
**double asterisks will produce strong character**
This is [an example](https://example.com/) inline link.
```

Up to 1024 characters (0/1024)

1

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Cancel
Previous
Next

Enter a name for the Alarm and click **next**.

Step 2: Configure actions

Edit

Actions
Notification
When in alarm, send a notification to "Hussain"

EC2 action
When in alarm, stop this instance (Instance ID: i-0af7e2dfee9b71a33)

Step 3: Add name and description

Edit

Name and description
Name
stop EC2 instance

Description
-

Cancel
Previous
Create alarm

Scroll down and click on create alarm

Alarms (1)

☐ Hide Auto Scaling alarms
Clear selection
Create composite alarm
Actions
Create alarm

☐	Name	State	Last state update (UTC)	Conditions	Actions
☐	stop EC2 instance	Insufficient data	2025-05-05 00:07:55	CPUUtilization < 10 for 1 datapoints within 5 minutes	Actions enabled

Wait for about 5 minutes for the alarm to be activated.

Note: the alarm that you create will check the CPU Utility and if it is below 10 it will stop the instance and will send you an E-Mail about it.

